Valentina Pucci

Neuro-X student at EPFL

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Location: Lausanne, CH

Presentation

I am a Neuro-X student at EPFL specializing in neuroengineering and computational methods, with a strong foundation in engineering, data analysis, and biomedical systems. Through my academic and project experience, I have worked on complex problems at the intersection of hardware, software, and medical technology. I am particularly interested in the development of medical devices and technologies that can improve patient care. Having played volleyball for most of my life, I have developed a strong appreciation for teamwork, discipline, and collaboration, which I bring into my work and into projects that combine engineering and neuroscience to advance human health.

Education

Master of Science | ***Neuro-X*** — September 2024 - Present
École polytechnique fédérale de Lausanne (EPFL) — Lausanne, CH

- Focus: Machine Learning, Data Analysis, Neural Signals and Signal Processing, Translational Neuroengineering, Neural Interfaces, Image Processing, Bioelectronics, Computational Motor Control
- Current GPA: **5.5** / 6

Bachelor of Science | ***Computer Science and Engineering*** — 2021 - 2024
Politecnico di Milano — Milano, IT

- Focus: Algorithms and Data Structures, Geometry and Linear Algebra, Probability and Statistics, Electronics
- Final grade: **110 with honors** / 110

Work Experience

Clinical Research Intern — Mar 2026 – Present
MYNERVA – MedTech Spin-off from ETH Zurich — Morges / Geneva, CH

- Contributing to a pilot clinical trial evaluating wearable neurotechnology for gait and balance assessment in patients with neuropathy.
- Collecting and analyzing patient data, supporting experimental troubleshooting, and ensuring data quality across study sessions.
- Contributing to clinical documentation, data interpretation, and internal reporting.

Teaching Assistant — Sept 2025 – Jan 2026
École polytechnique fédérale de Lausanne (EPFL) — Lausanne, CH

- Teaching Assistant for the Neural Signals and Signal Processing course.
- Supported students during tutorials and practical sessions and provided guidance on course projects.

Peer to Peer Tutor — Oct 2022 – Mar 2023
Politecnico di Milano — Milano, IT

- Provided tutoring to university peers in Geometry and Linear Algebra.

Volunteering

Neuro Student NetworkX Association Member – Events and Conferences — 2024 – Present
École polytechnique fédérale de Lausanne (EPFL) — Lausanne, CH

- Organizing academic and networking events for students and professionals in neuroengineering.
- Contributing to the organization of conferences and outreach to industry and academic partners.

Volunteer Homework Help — 2017 - 2020
Le Ali di Chiara — Rimini, IT

- Provided academic assistance and language support to elementary and middle school students from diverse cultural backgrounds.

Relevant Projects and Research

Characterization of kirigami-based devices for long-term implantation Sept 2025 - Feb 2026
Laboratory of Soft Bioelectronic Interfaces Geneva, CH

- Investigating the electrochemical stability of kirigami-inspired soft implantable electrodes for chronic neural interfacing through ageing studies, focusing on impedance evolution and material degradation mechanisms

Hybrid Computational Models of Thoracic Spinal Cord Stimulation Feb - Aug 2025
Translational Neuroengineering Lab, REHAssist and MINE Lab Geneva (CH), Lausanne (CH) and Milano (IT)

- Developed a Hybrid Computational Model of the Spinal Cord and stimulation system with Transcutaneous and Epidural electrodes. Integrated MATLAB and Python coding, COMSOL Multiphysics for modeling, and NEURON for physical simulations to build an open source toolbox for analyzing recruitment and selectivity of spinal circuits due to different stimulation patterns. Collaborated directly with clinicians and patients at IRCCS San Raffaele Hospital (Milan) to ensure translational relevance of the models.

Task Prediction Using fMRI Data Nov - Dec 2024
École polytechnique fédérale de Lausanne (EPFL) Lausanne, CH

- Developed Recurrent Neural Network models to predict the timecourses of task paradigms using brain imaging data (fMRI) from the Human Connectome Project's task-based fMRI dataset.

Honors and Awards

Lead The Future Mentorship 2024 - Present
Italy IT

- Among the few Italian students selected to be mentees for LeadTheFuture, a leading mentorship non-profit organization for students in STEM, with acceptance rate below 20%. https://www.leadthefuture.tech

Best Freshman Award 2021-2022
Politecnico di Milano Milano, IT

- Granted from Politecnico di Milano a **Merit Scholarship** for the Academic Year 2021-2022 as I was among the highest achieving students in my department.

Skills

Languages:

- **Italian:** Native
- **English:** Full proficiency
- **French:** B1 level

Programming:

- **Python:** Advanced
- **MATLAB:** Advanced
- **C:** Advanced
- **Java:** Intermediate
- **COMSOL Multiphysics**: Intermediate
- **HTML:** Intermediate
- **JavaScript:** Intermediate
- **VHDL:** Intermediate